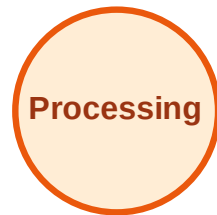


BFS Traversal

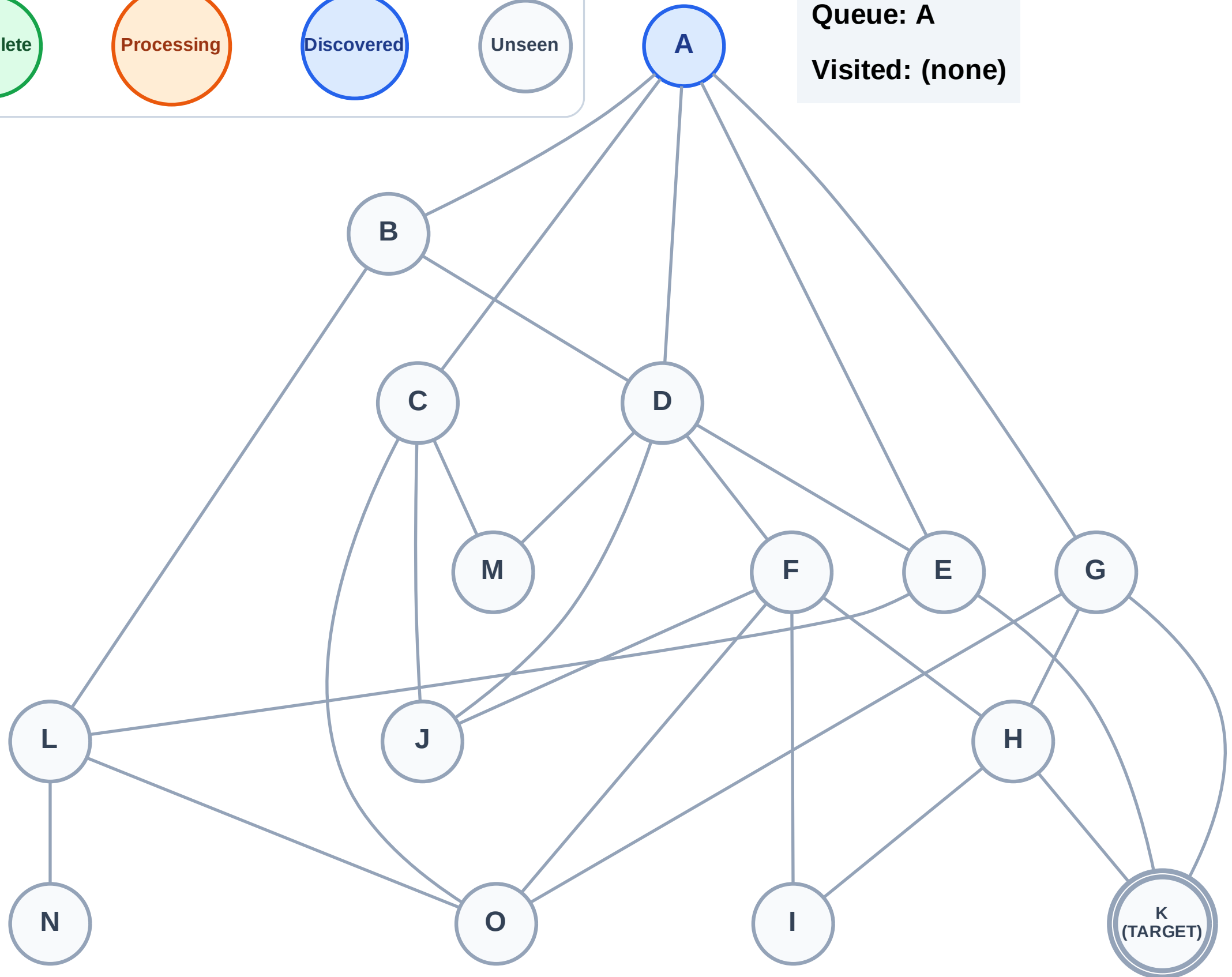
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

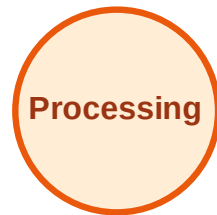
Visited: (none)



BFS Traversal

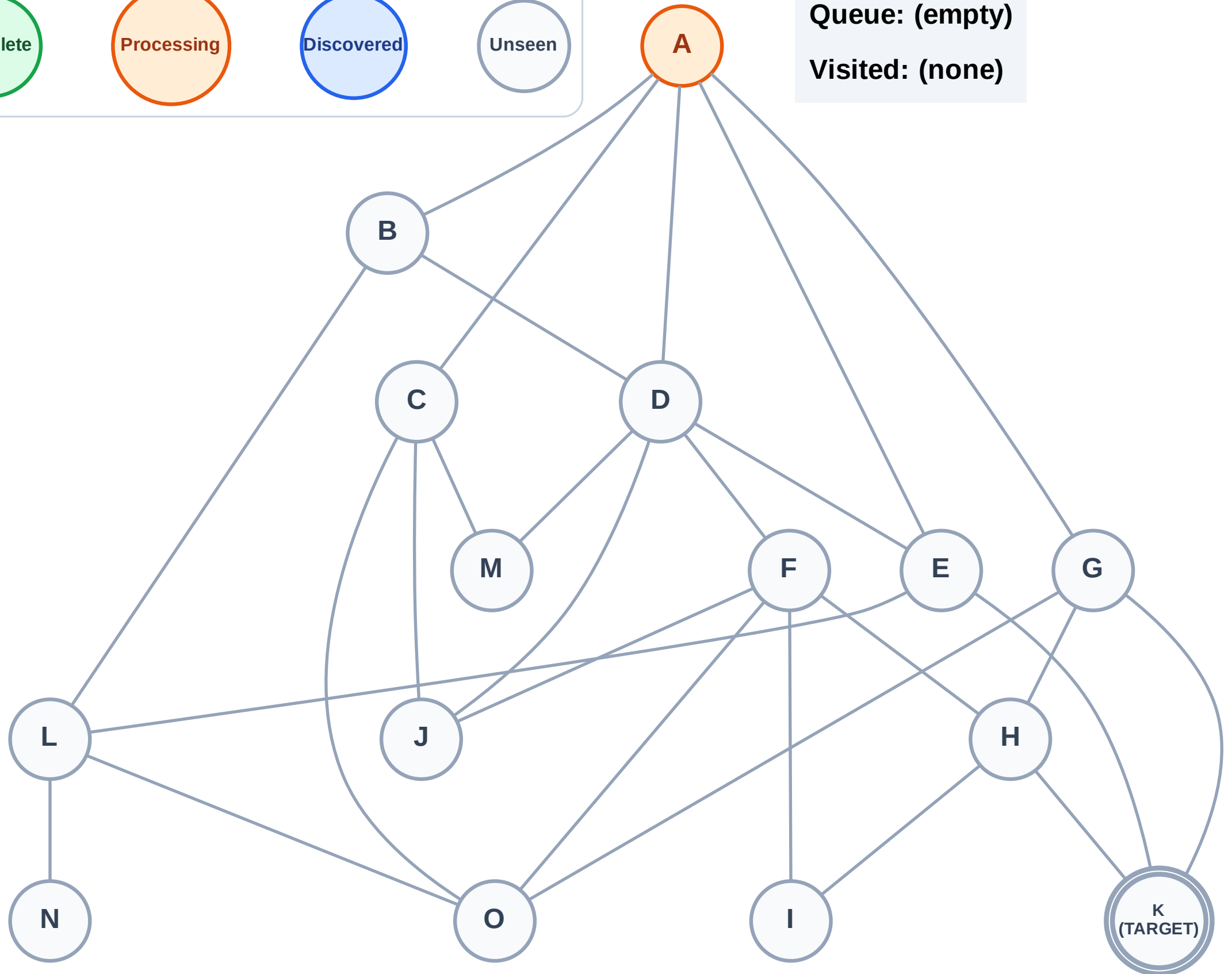
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, C, D, E, G from A

Legend

Complete

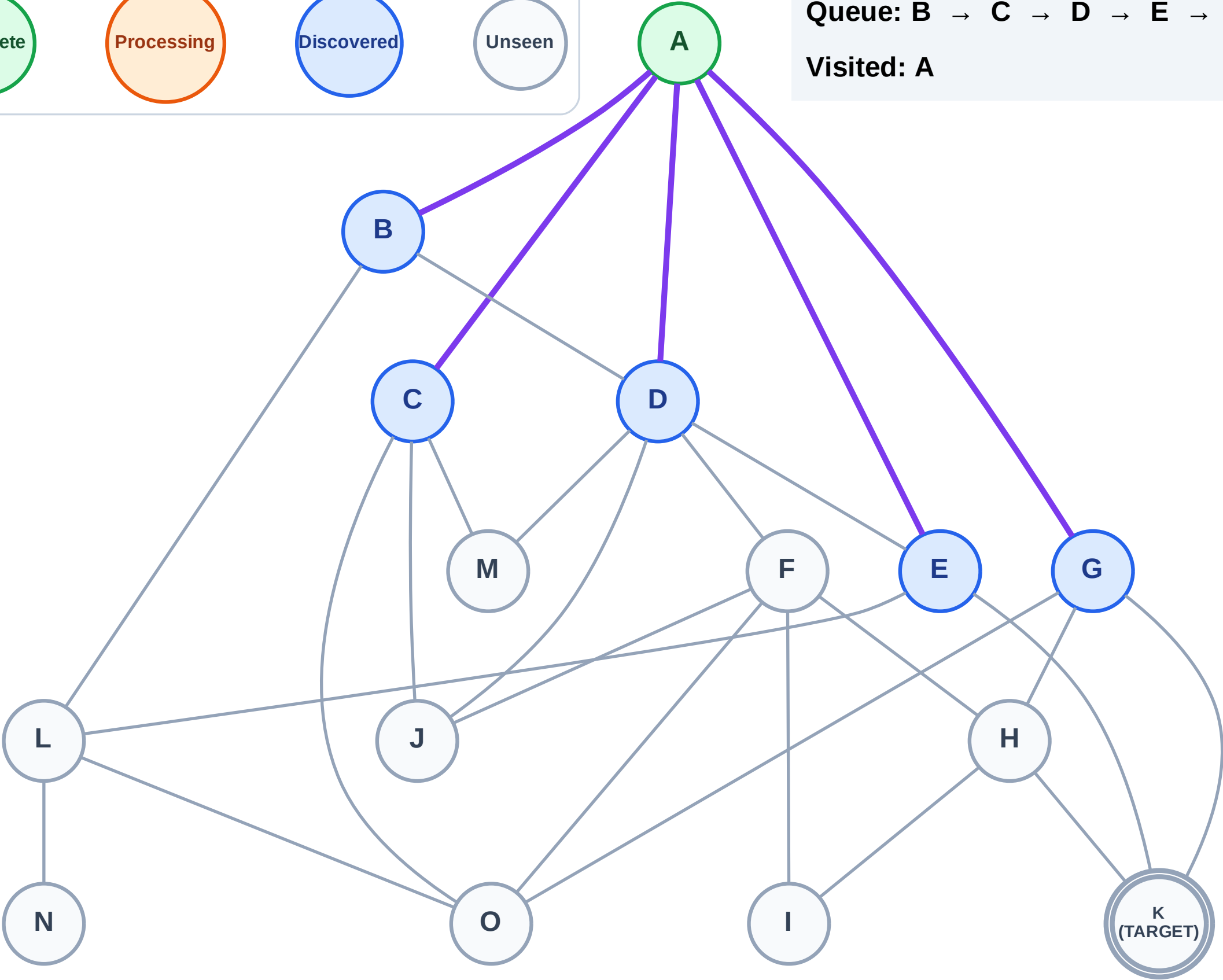
Processing

Discovered

Unseen

Queue: B → C → D → E → G

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

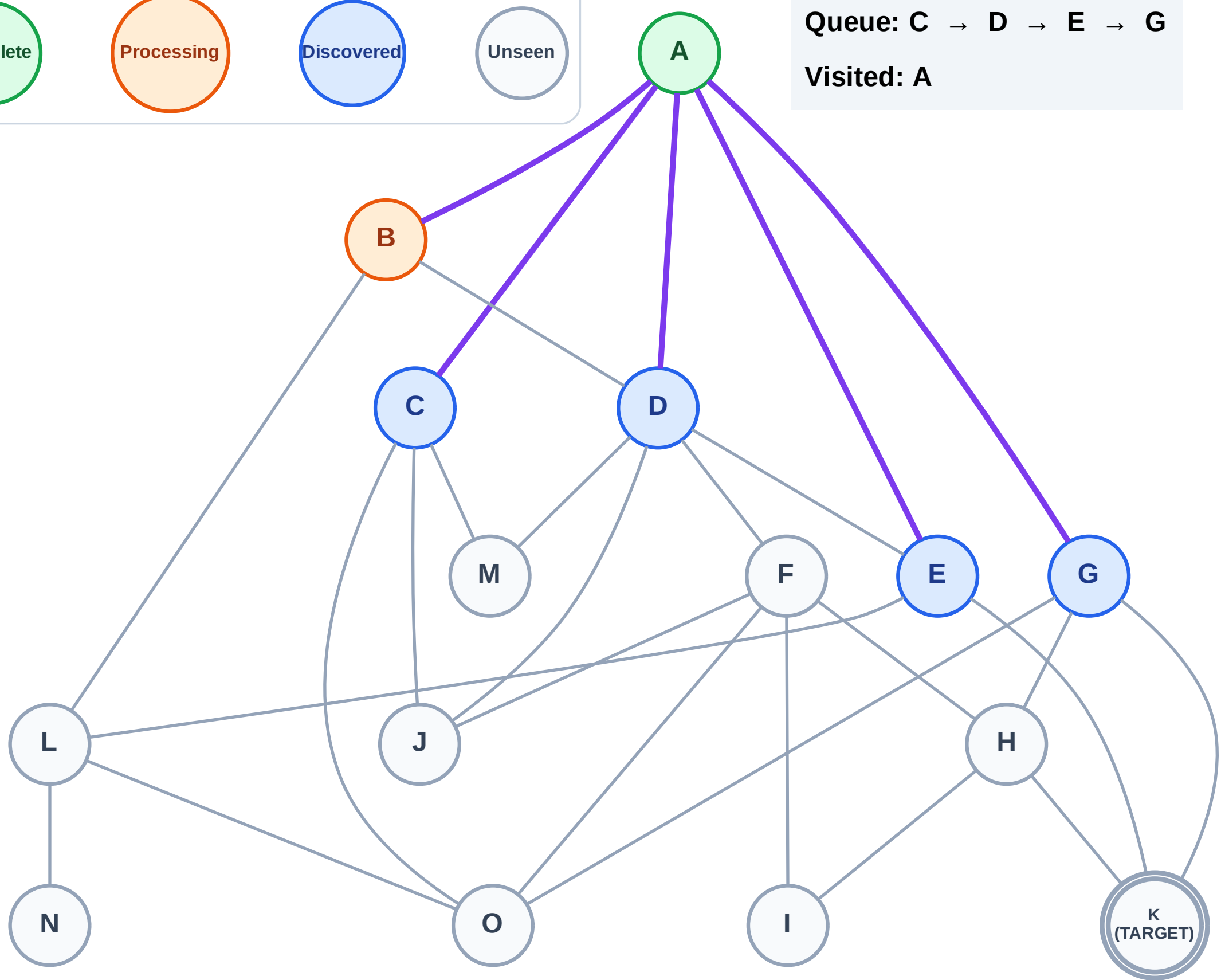
Processing

Discovered

Unseen

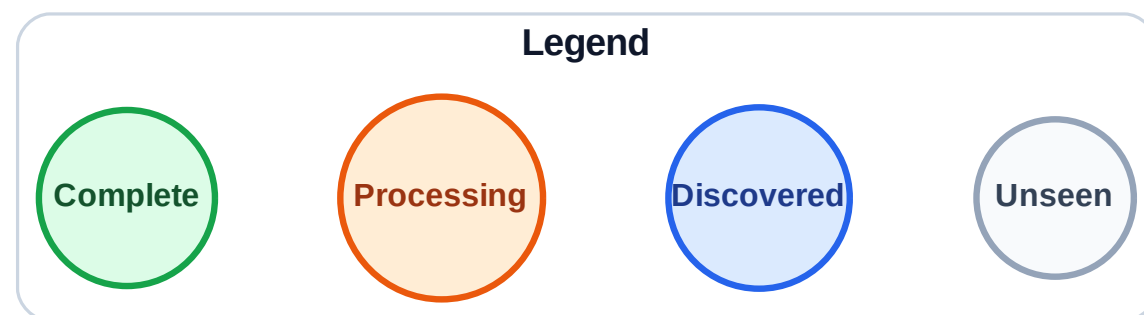
Queue: C → D → E → G

Visited: A



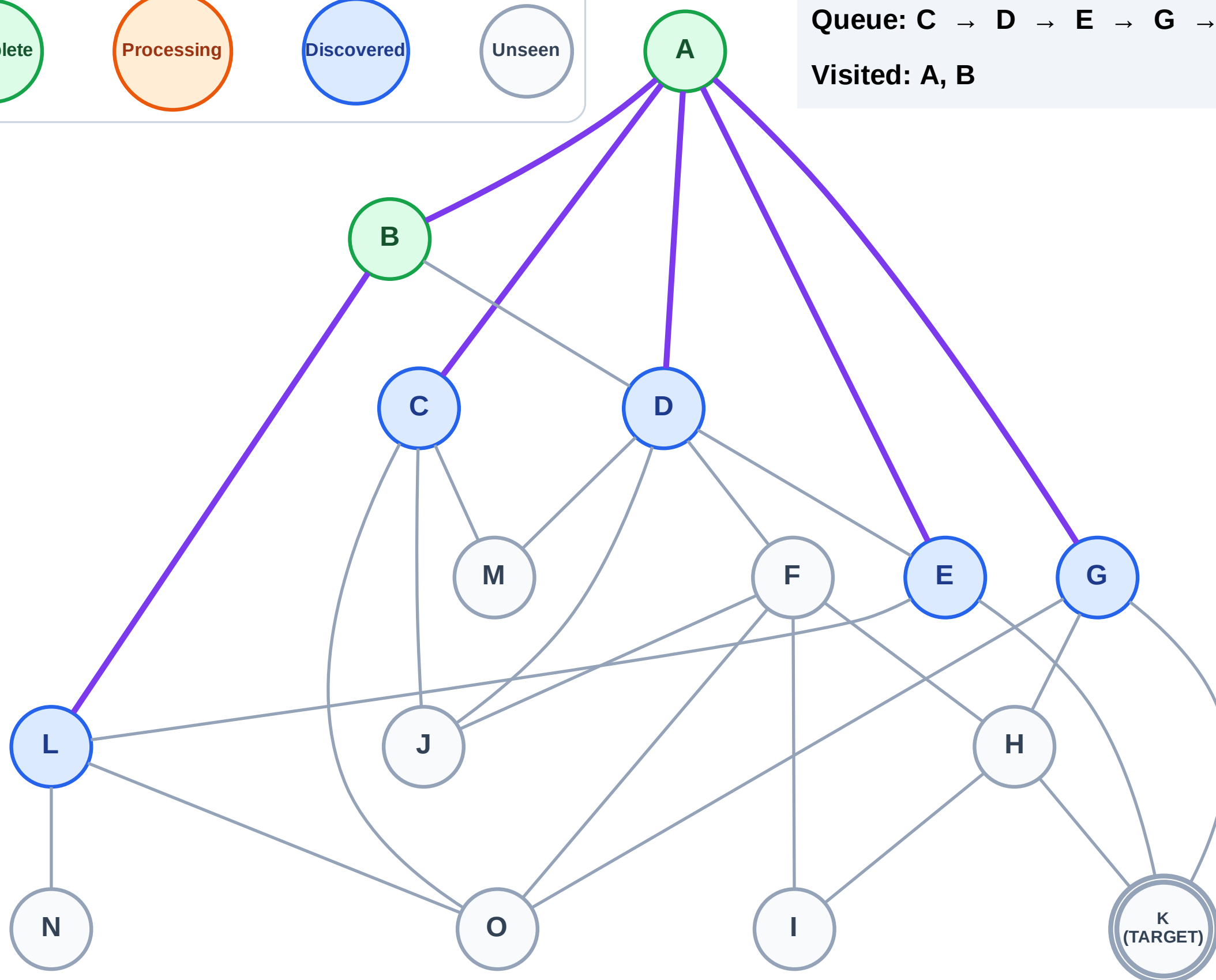
BFS Traversal

Step 5: Discover L from B



Queue: C → D → E → G → L

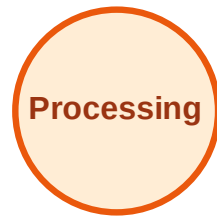
Visited: A, B



BFS Traversal

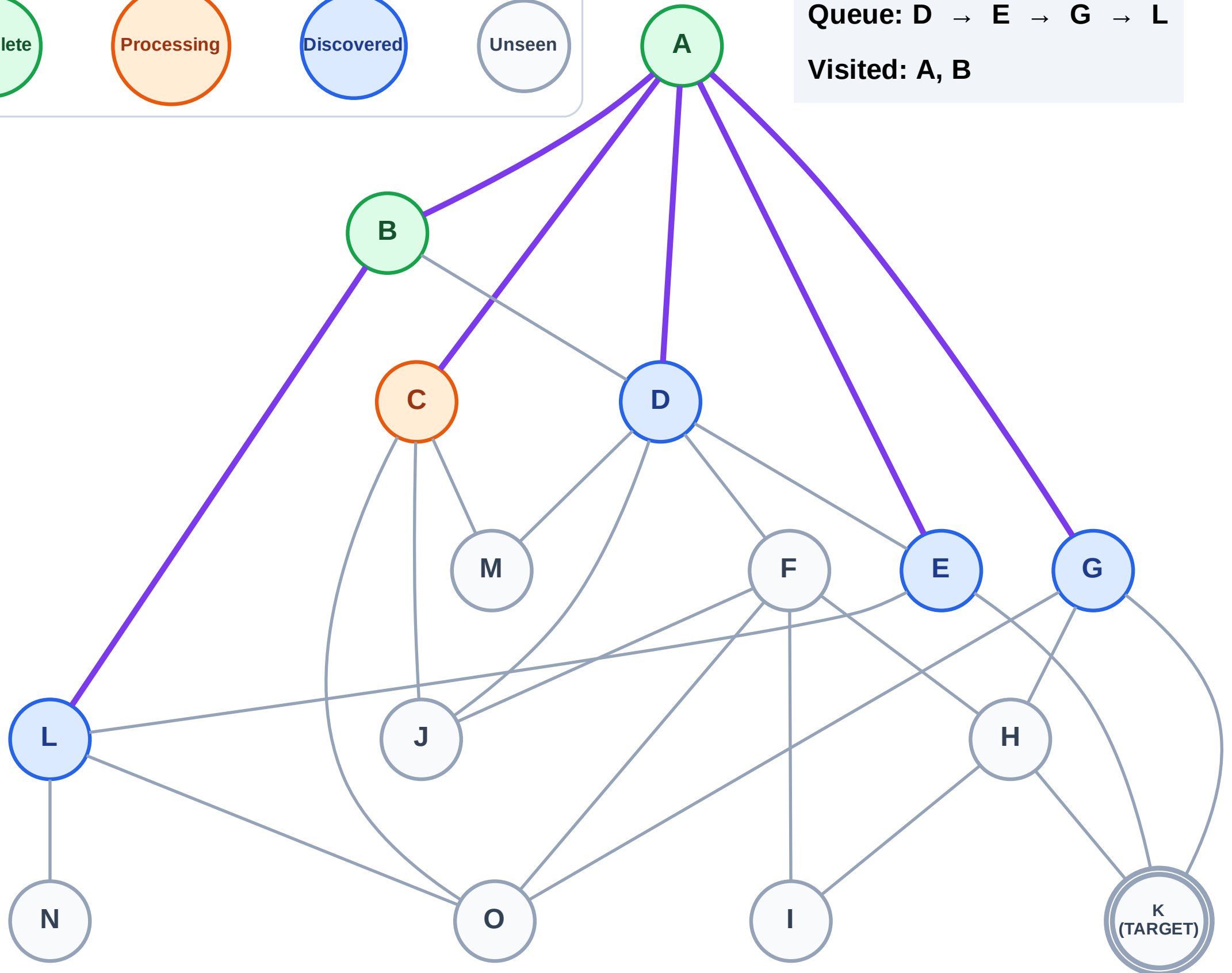
Step 6: Process node C

Legend



Queue: D → E → G → L

Visited: A, B



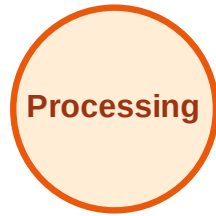
BFS Traversal

Step 7: Discover J, M, O from C

Legend



Complete



Processing



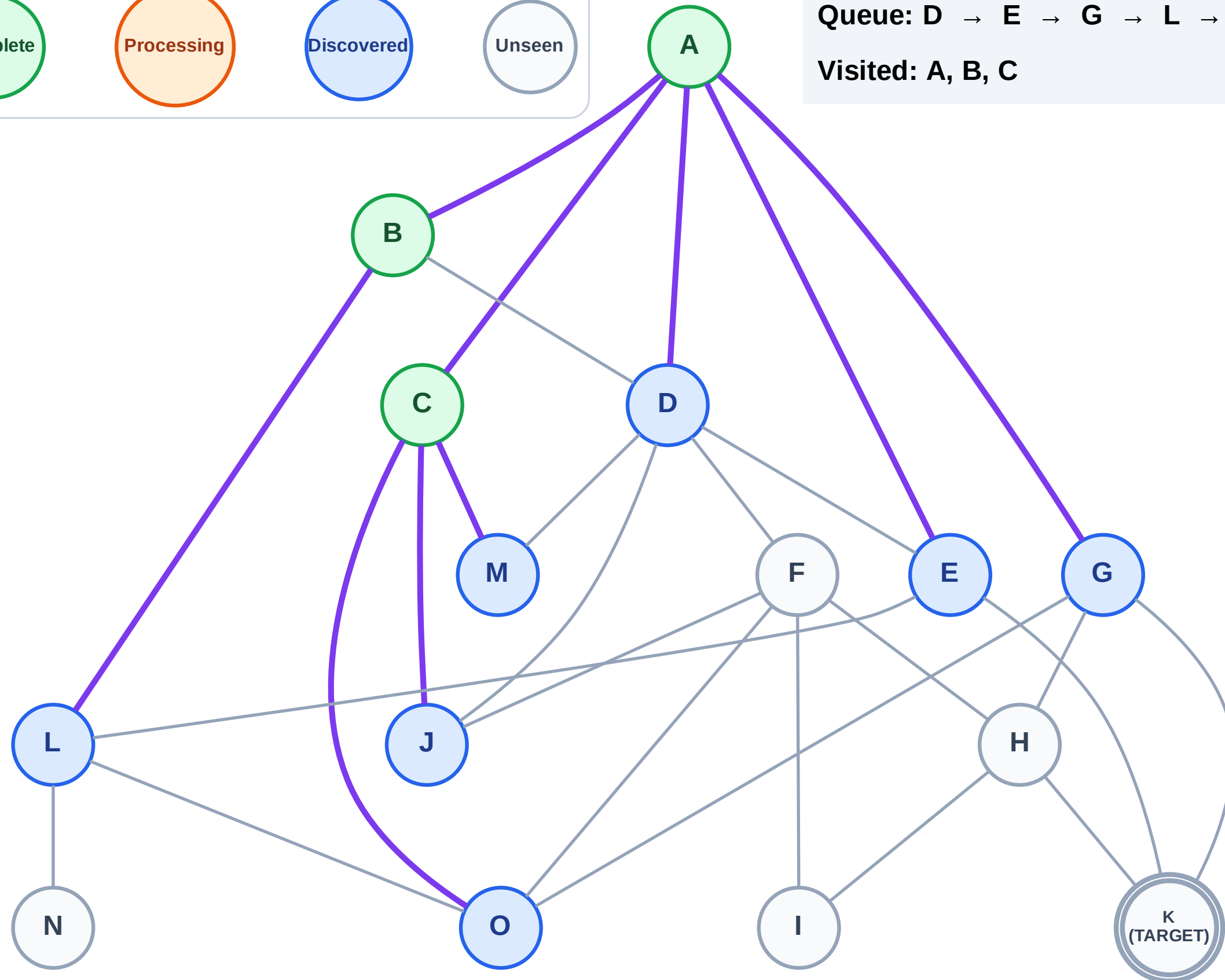
Discovered



Unseen

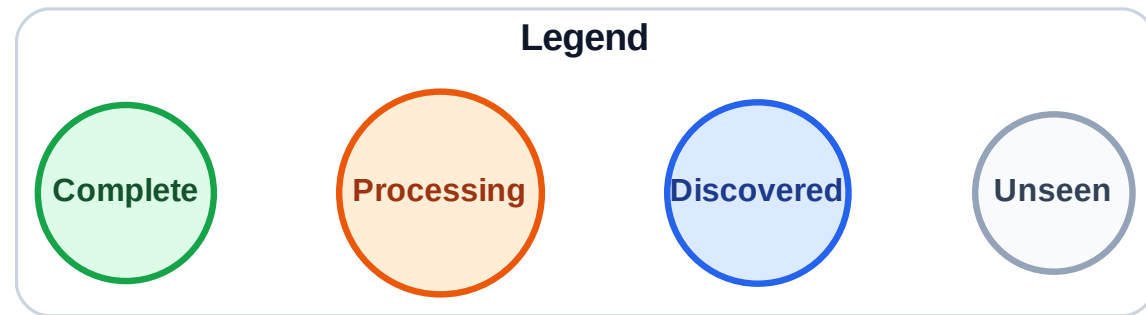
Queue: D → E → G → L → J → M → O

Visited: A, B, C



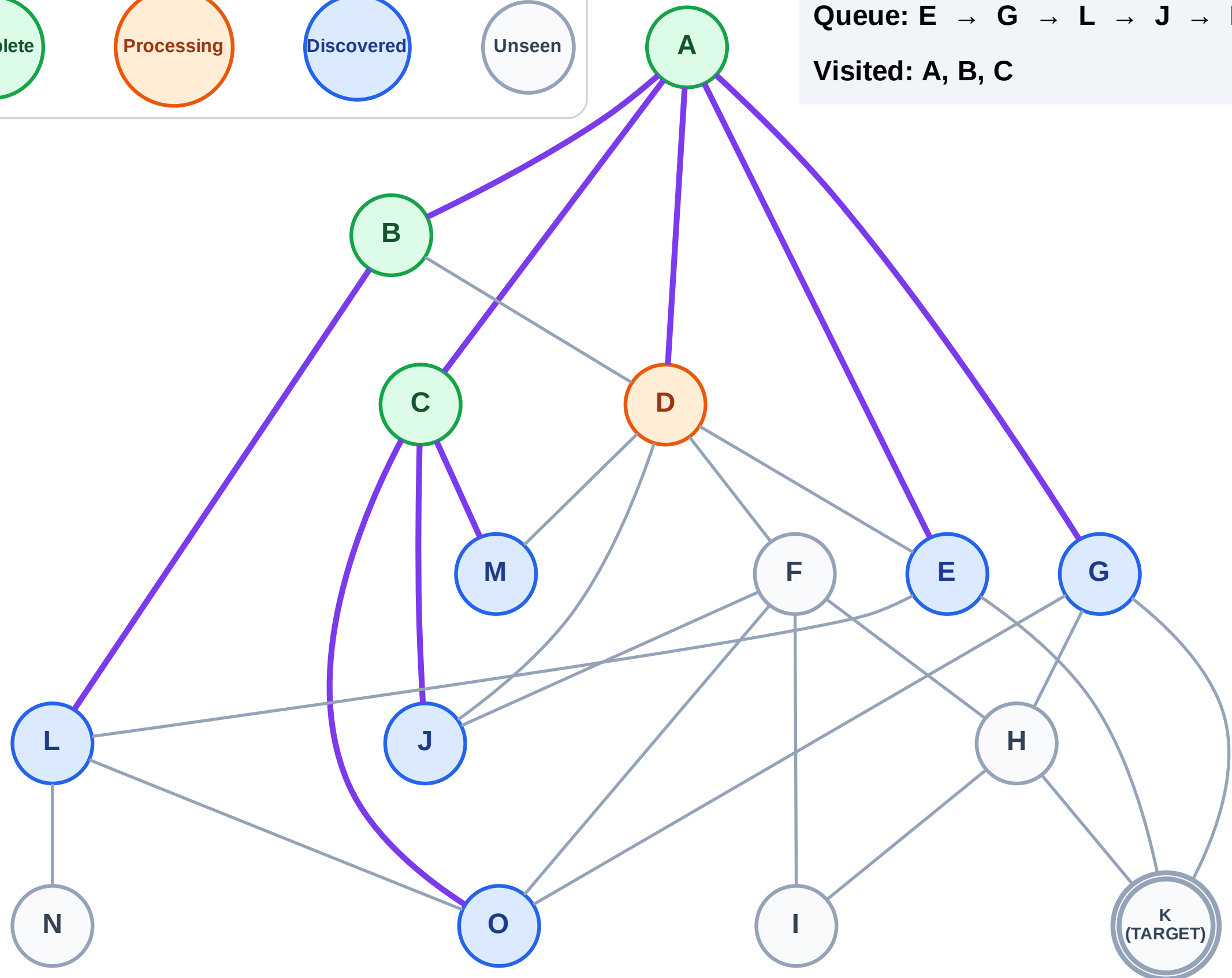
BFS Traversal

Step 8: Process node D



Queue: E → G → L → J → M → O

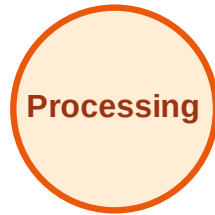
Visited: A, B, C



BFS Traversal

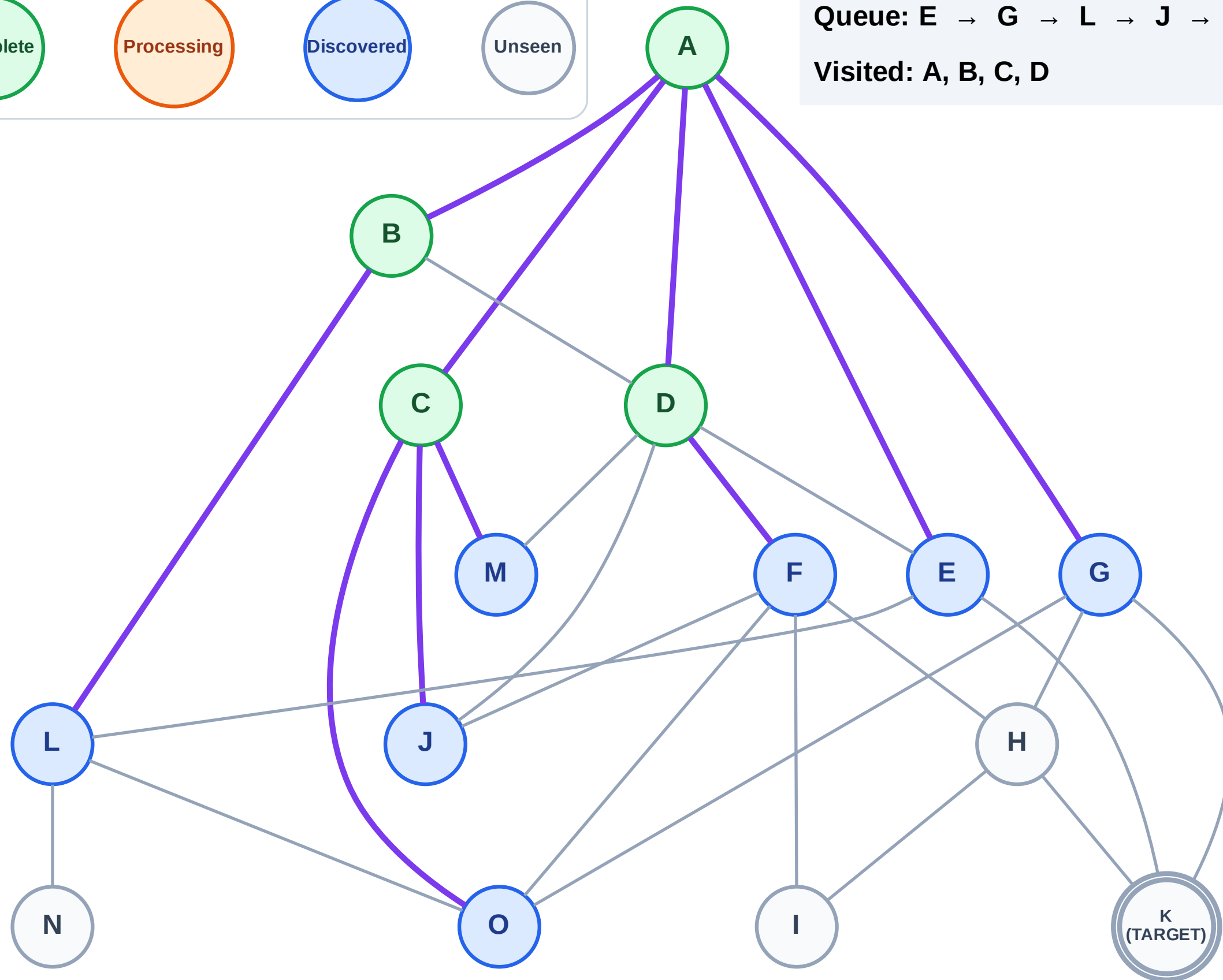
Step 9: Discover F from D

Legend



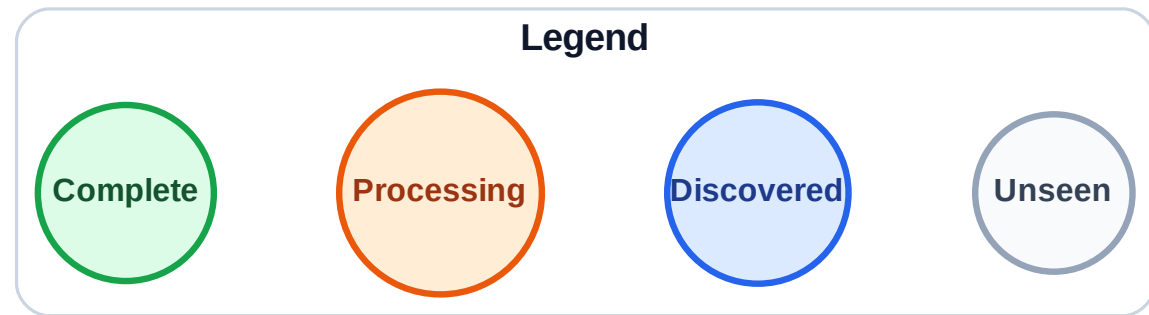
Queue: E → G → L → J → M → O → F

Visited: A, B, C, D



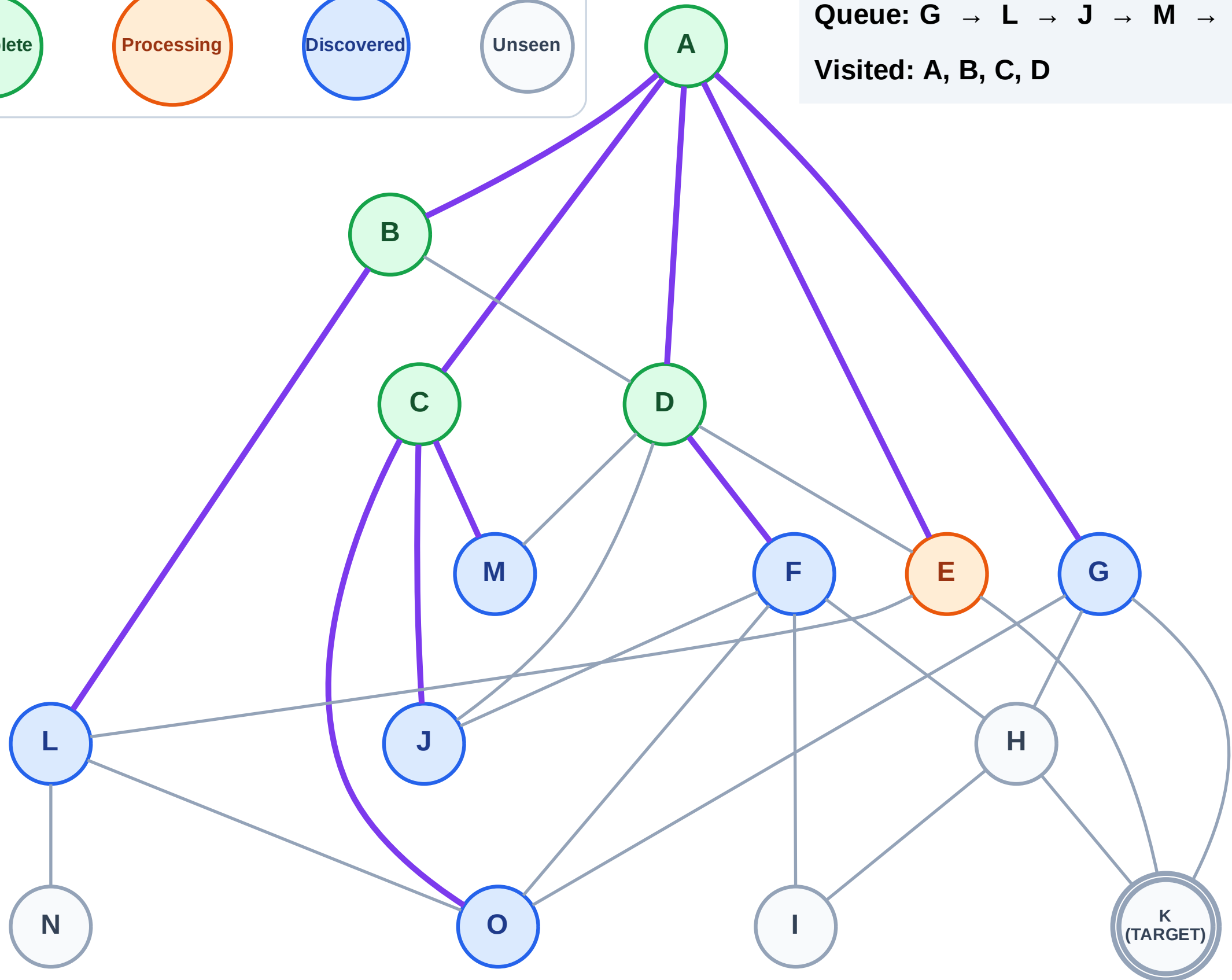
BFS Traversal

Step 10: Process node E



Queue: G → L → J → M → O → F

Visited: A, B, C, D



BFS Traversal

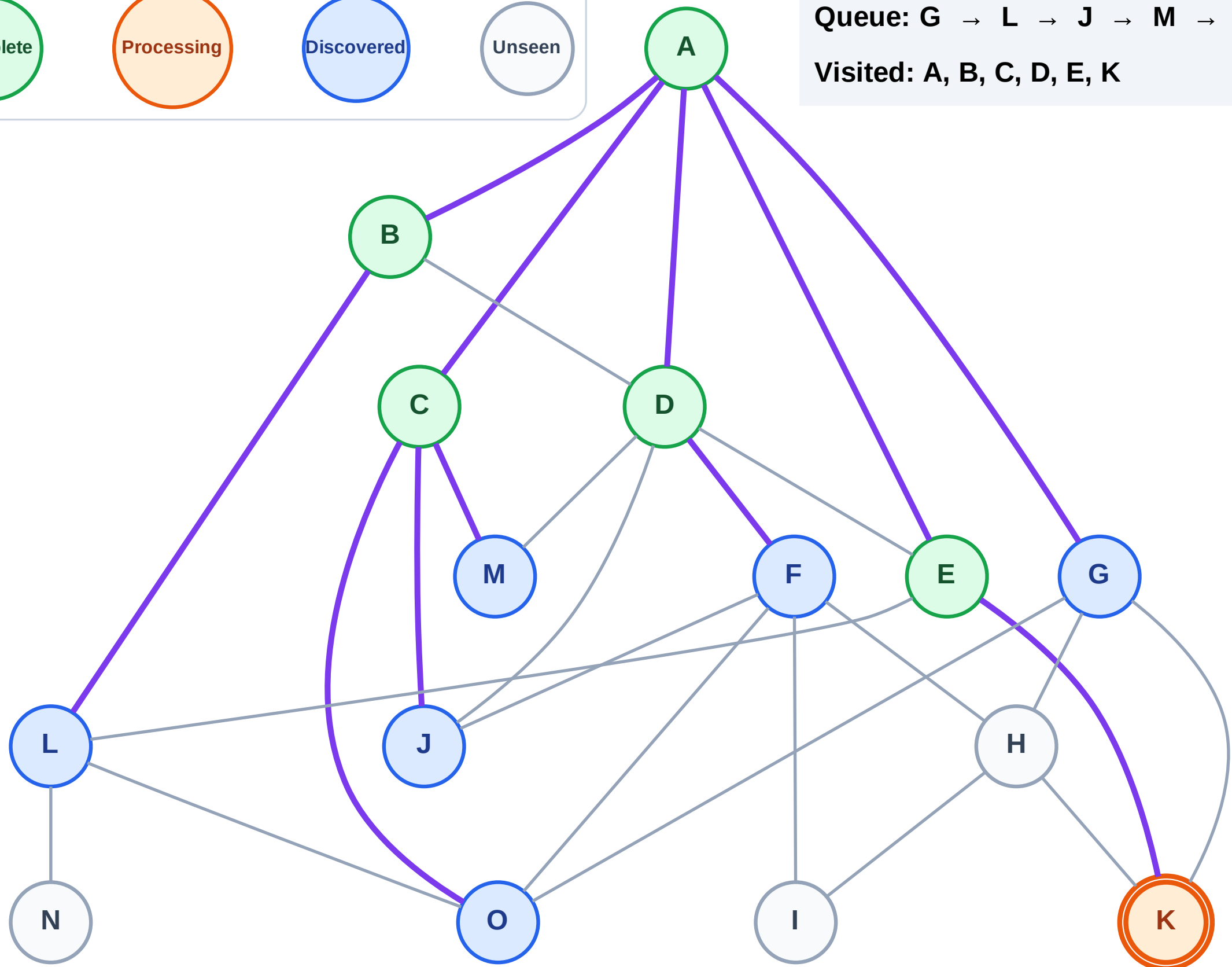
Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → E → K

Legend



Queue: G → L → J → M → O → F

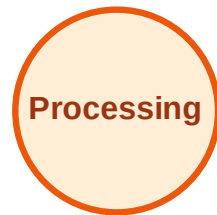
Visited: A, B, C, D, E, K



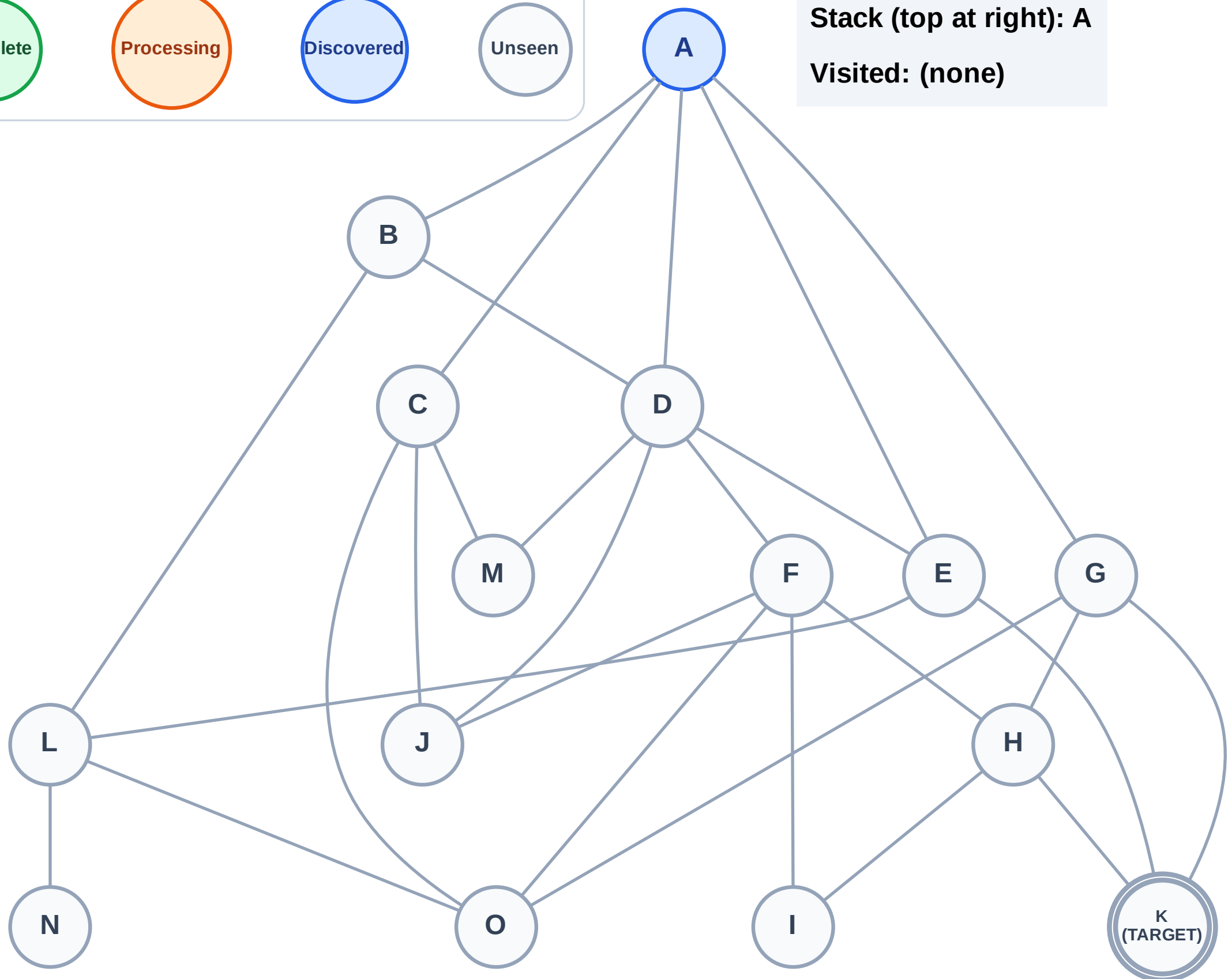
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



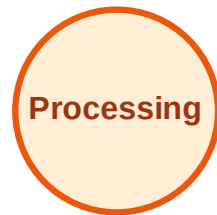
Stack (top at right): A
Visited: (none)



DFS Traversal

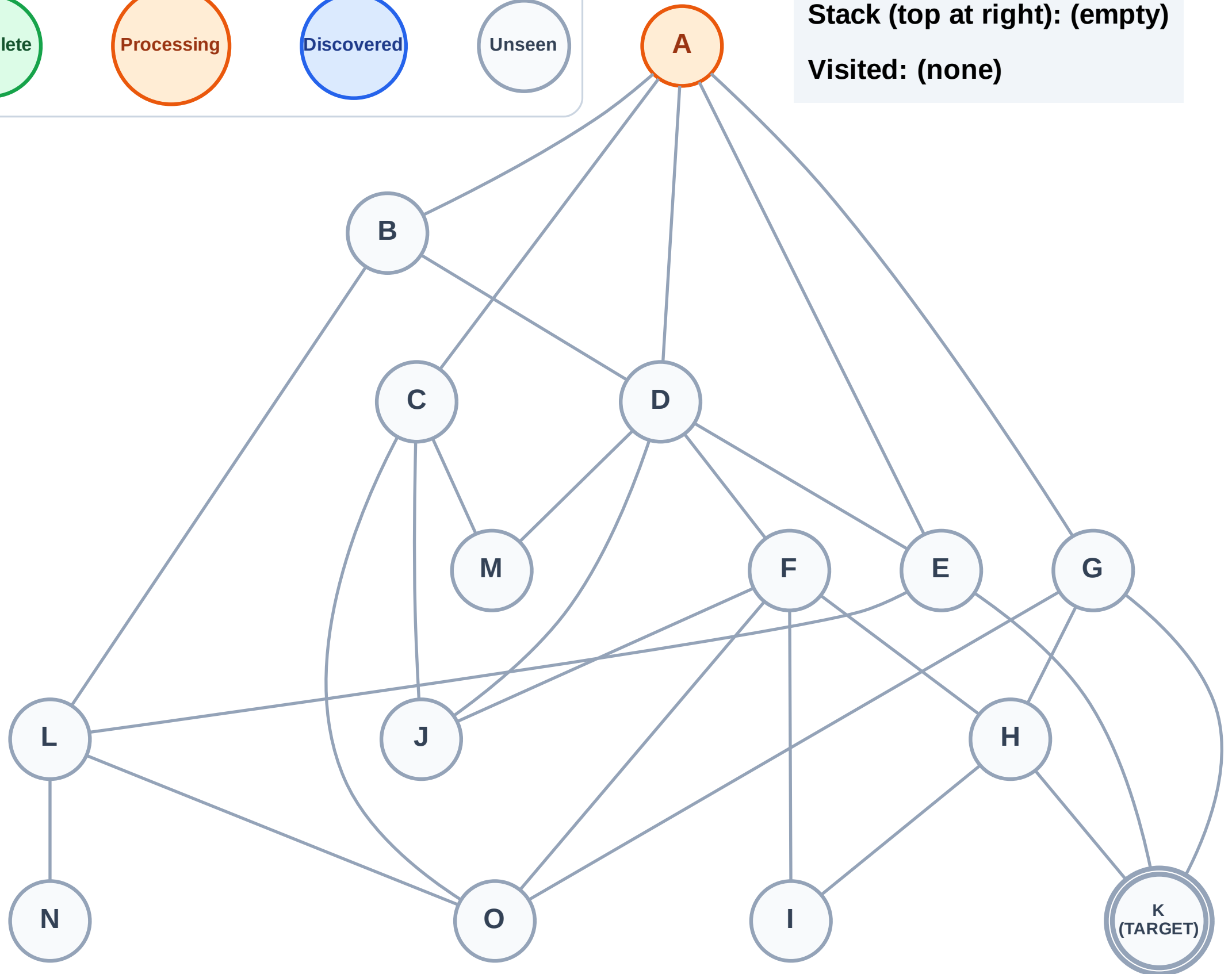
Step 2: Process node A

Legend



Stack (top at right): (empty)

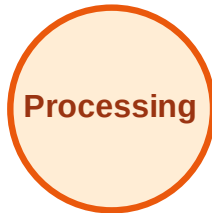
Visited: (none)



DFS Traversal

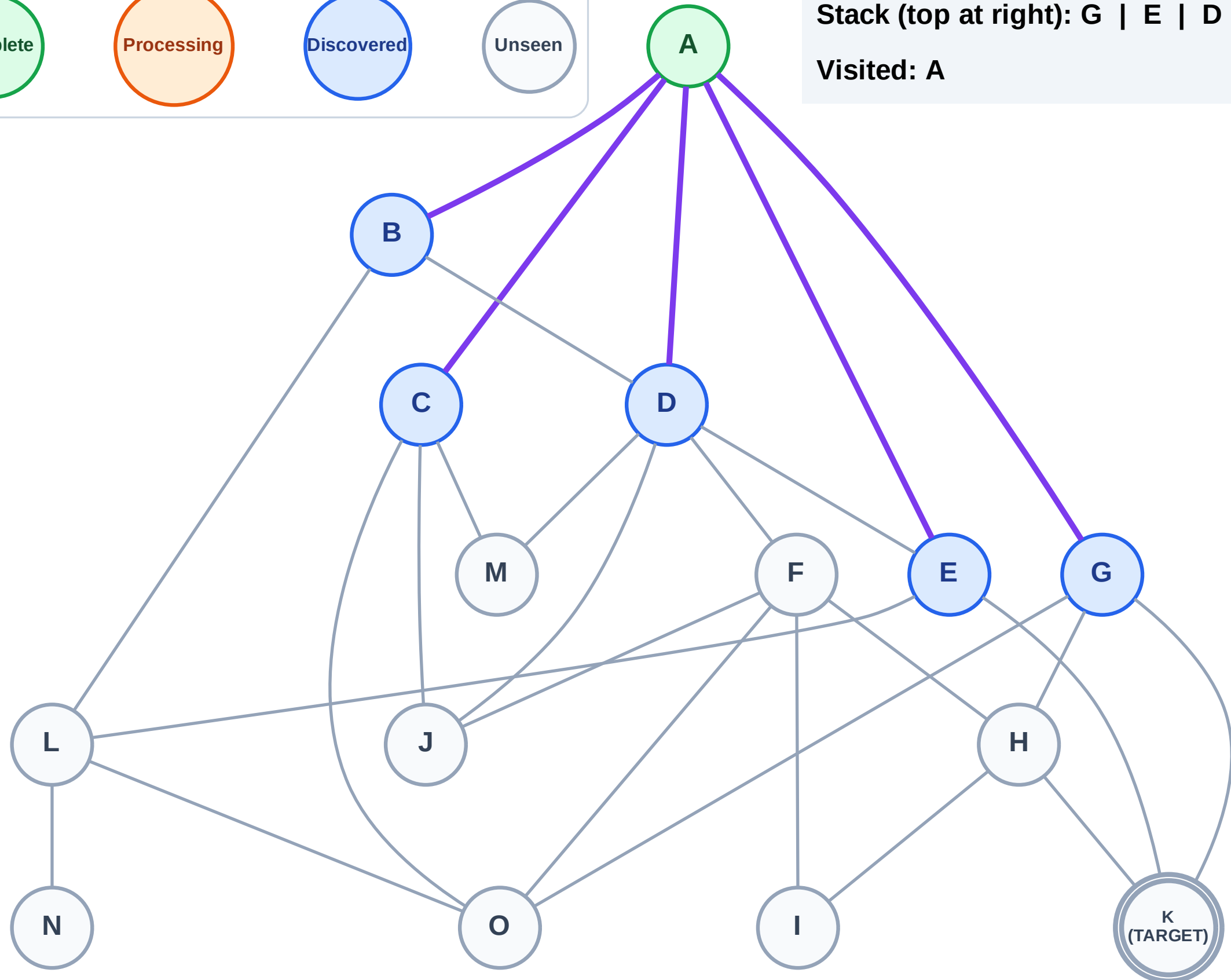
Step 3: Discover G, E, D, C, B from A

Legend



Stack (top at right): G | E | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

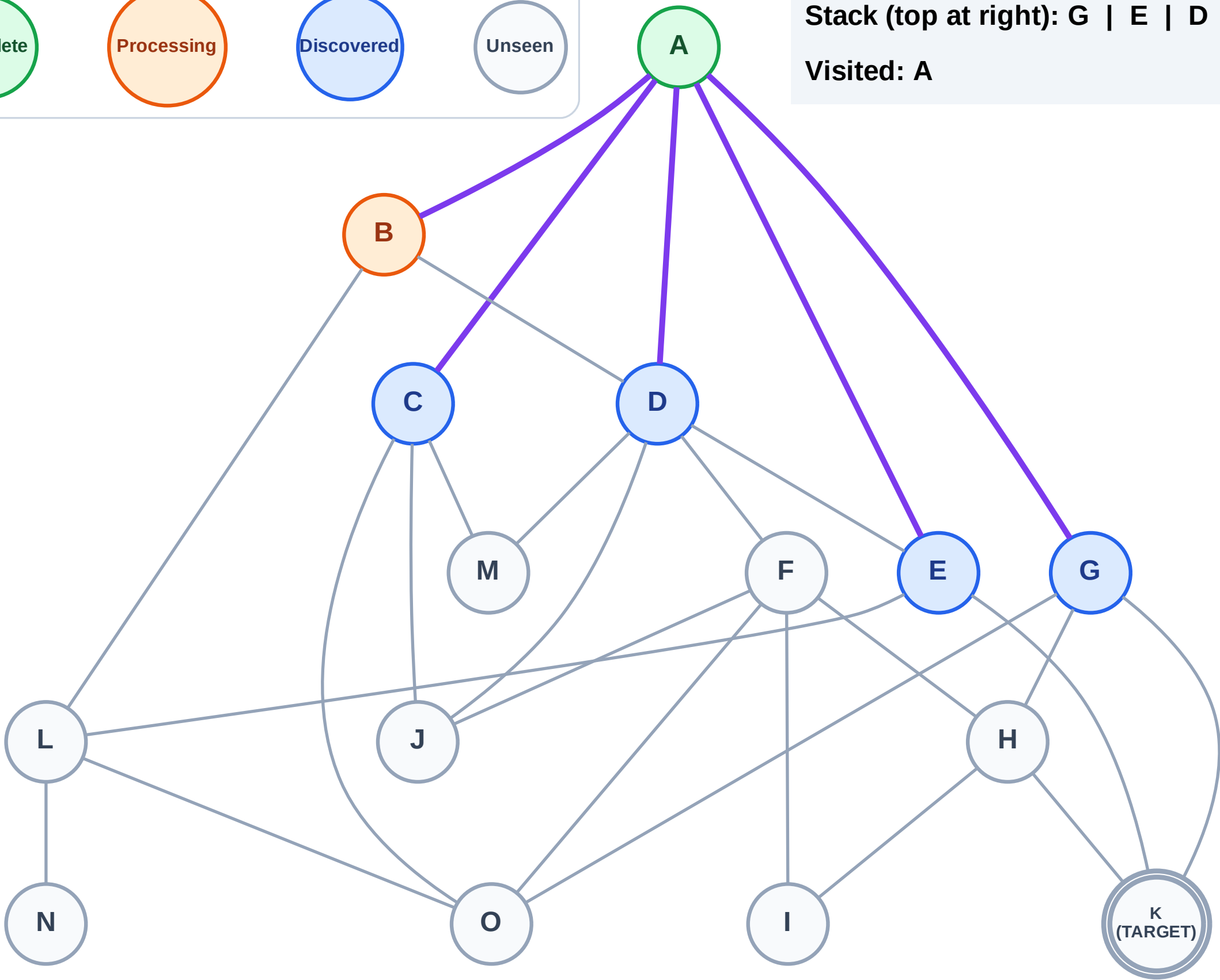
Processing

Discovered

Unseen

Stack (top at right): G | E | D | C

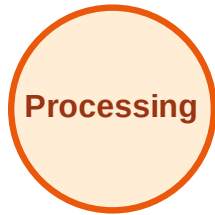
Visited: A



DFS Traversal

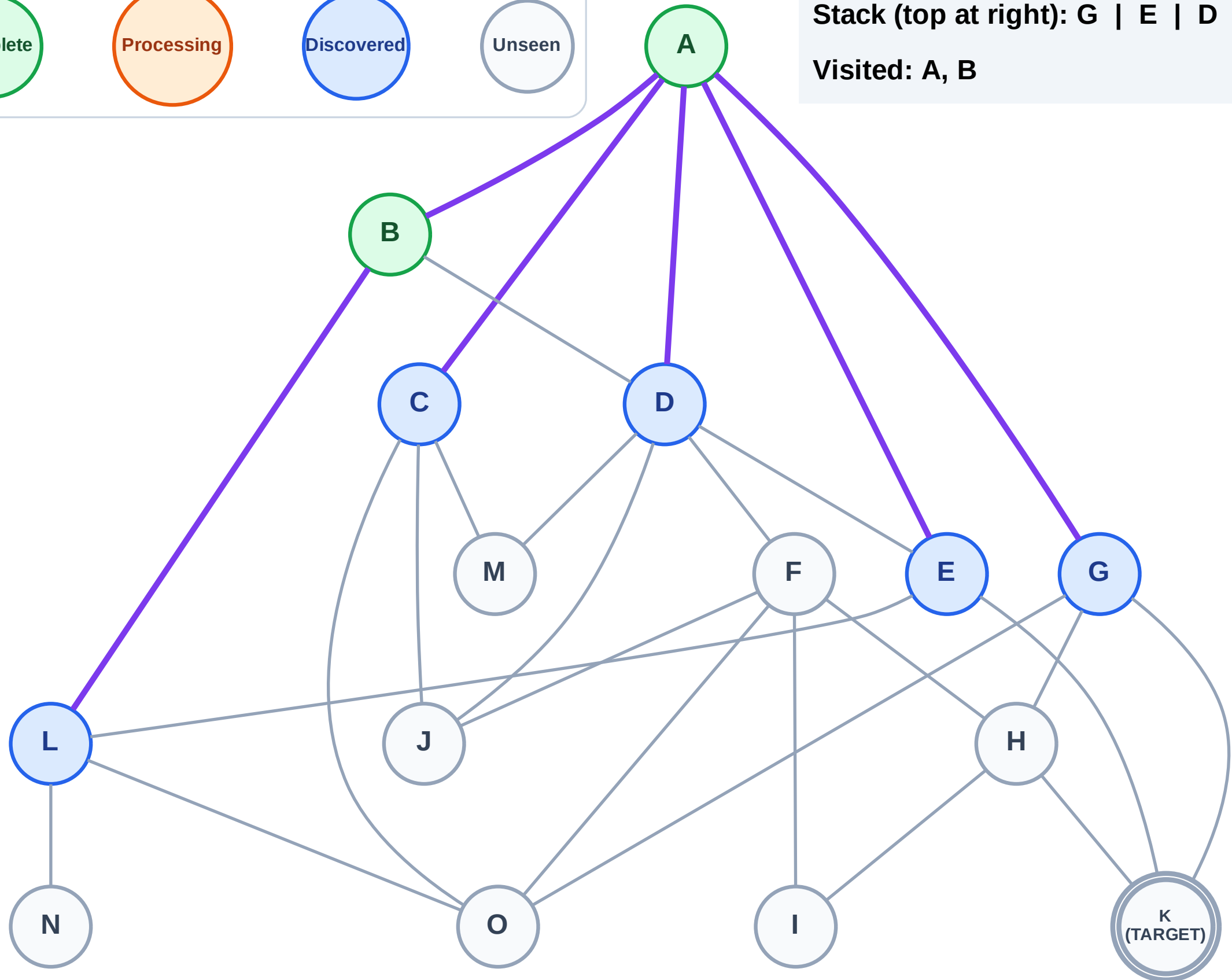
Step 5: Discover L from B

Legend



Stack (top at right): G | E | D | C | L

Visited: A, B



DFS Traversal

Step 6: Process node L

Legend

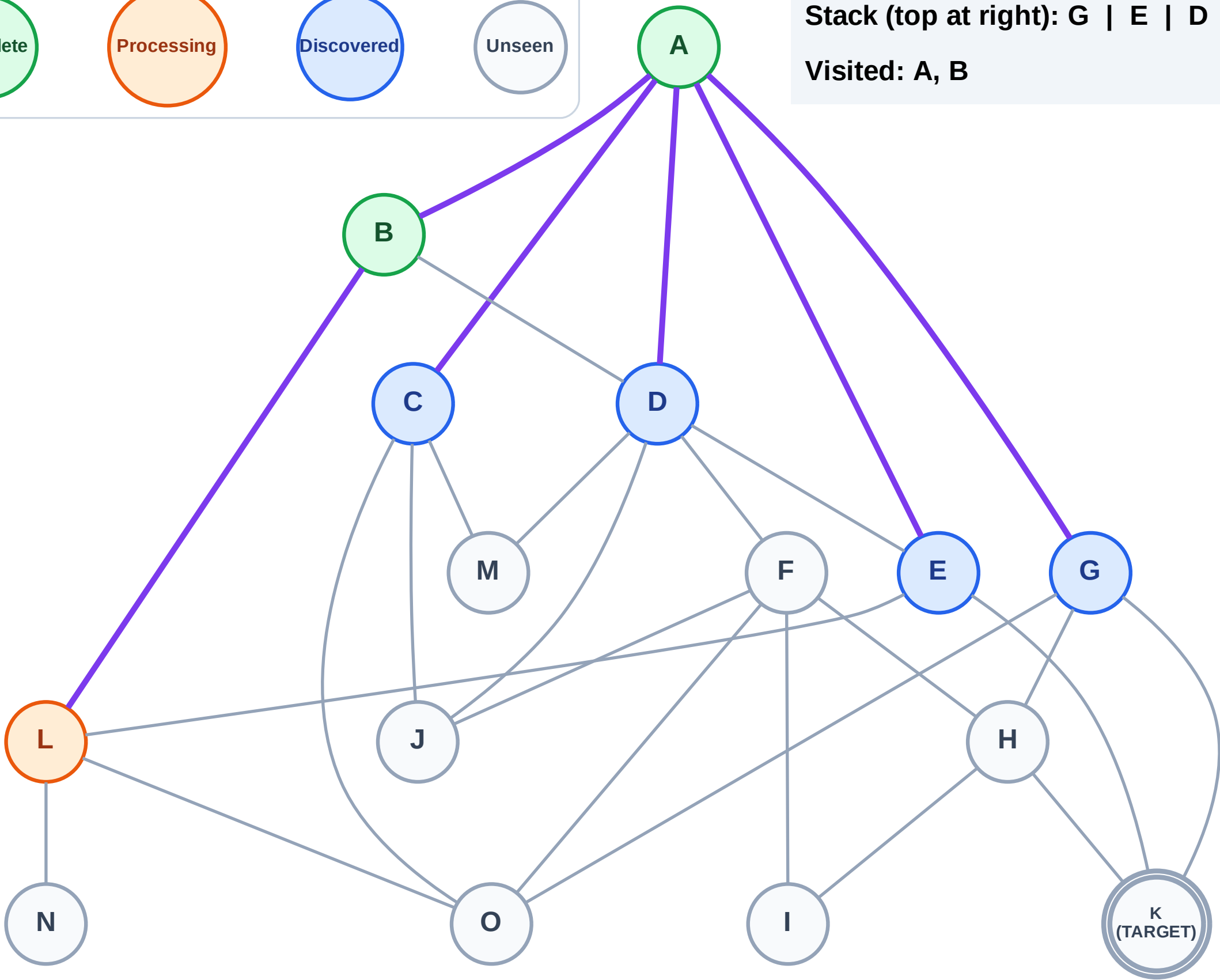
Complete

Processing

Discovered

Unseen

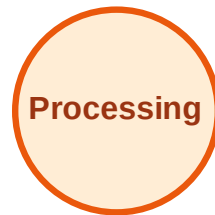
Stack (top at right): G | E | D | C
Visited: A, B



DFS Traversal

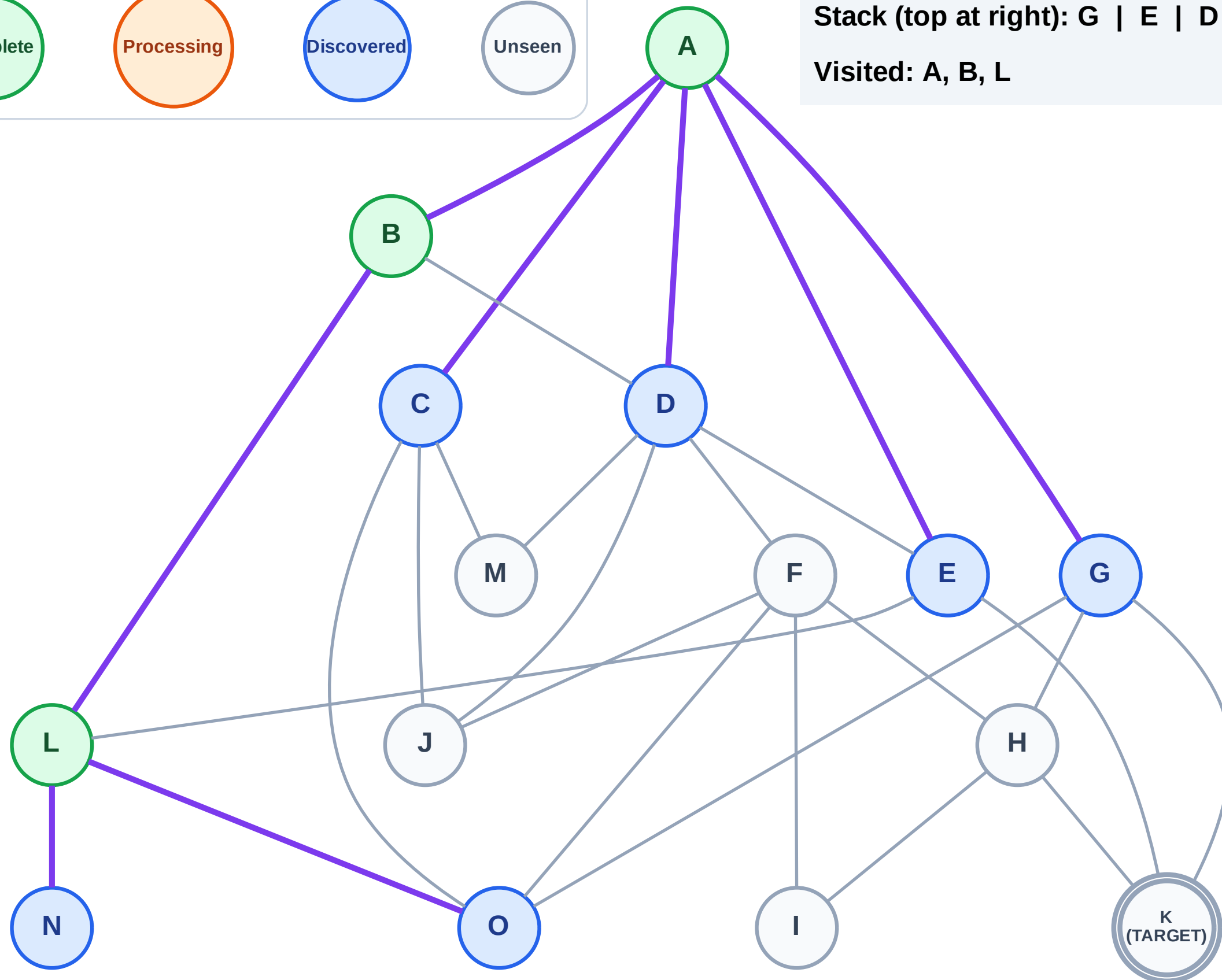
Step 7: Discover O, N from L

Legend



Stack (top at right): G | E | D | C | O | N

Visited: A, B, L



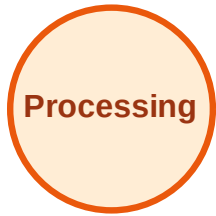
DFS Traversal

Step 8: Process node N

Legend



Complete



Processing



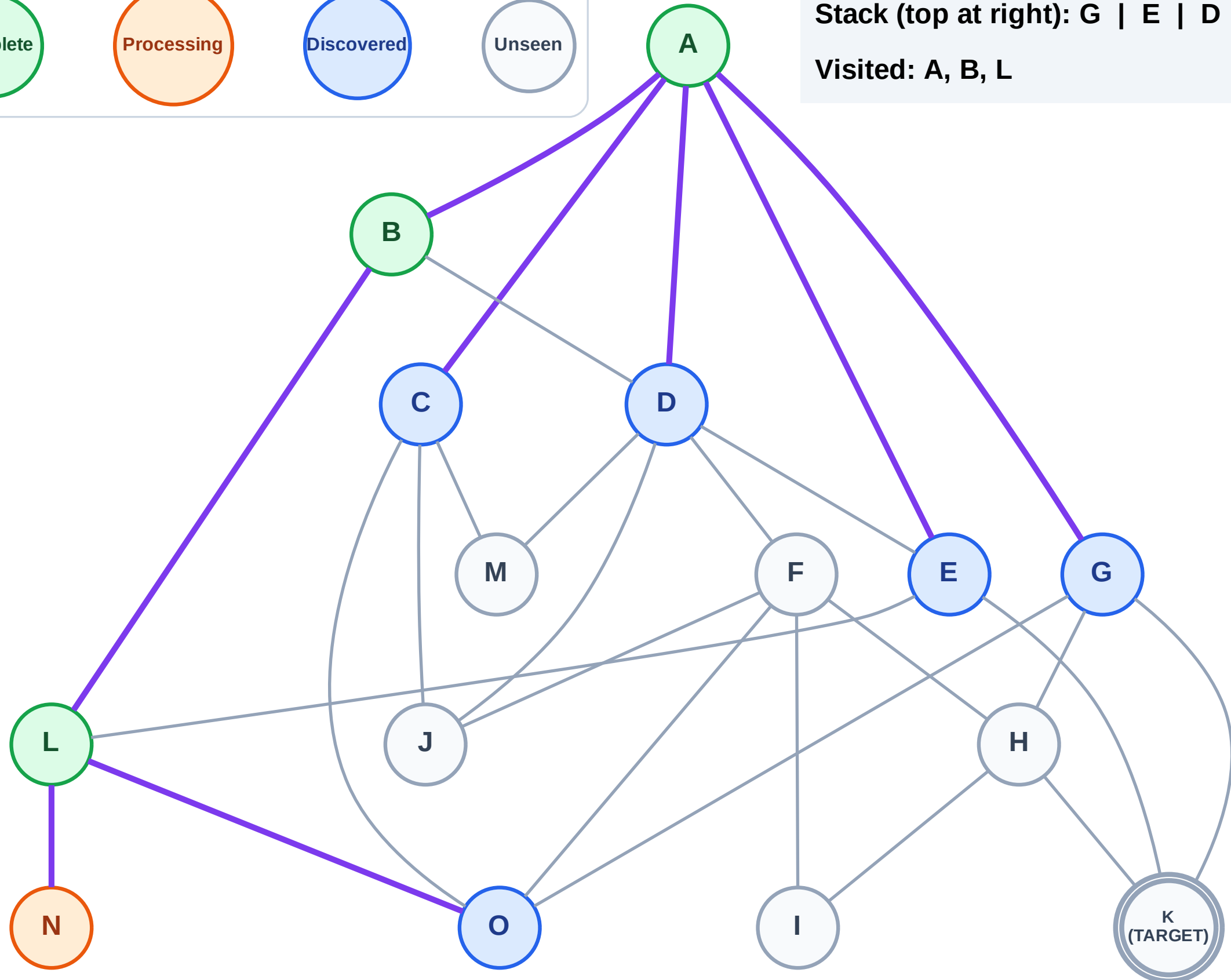
Discovered



Unseen

Stack (top at right): G | E | D | C | O

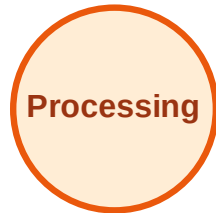
Visited: A, B, L



DFS Traversal

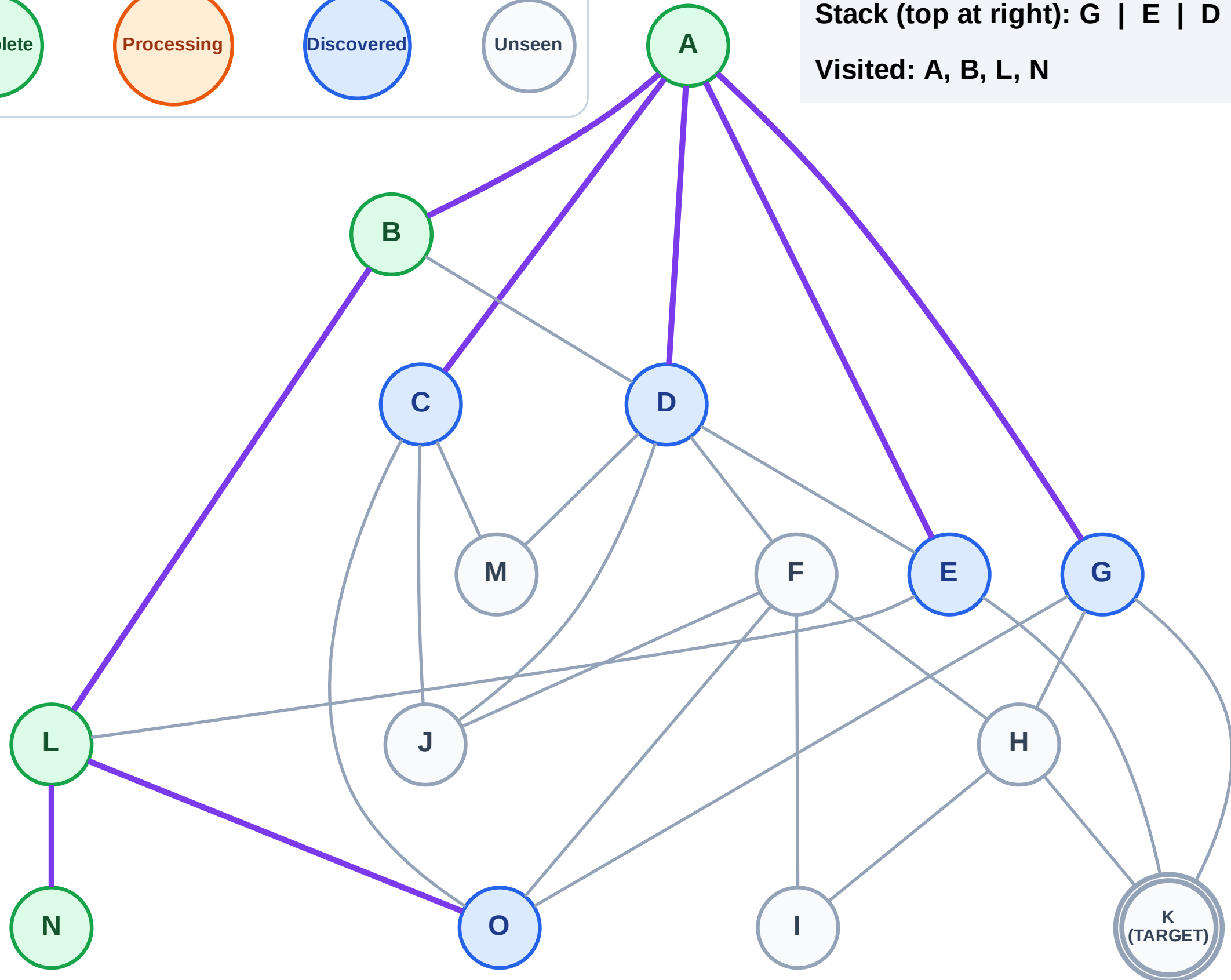
Step 9: No new neighbors from N

Legend



Stack (top at right): G | E | D | C | O

Visited: A, B, L, N



DFS Traversal

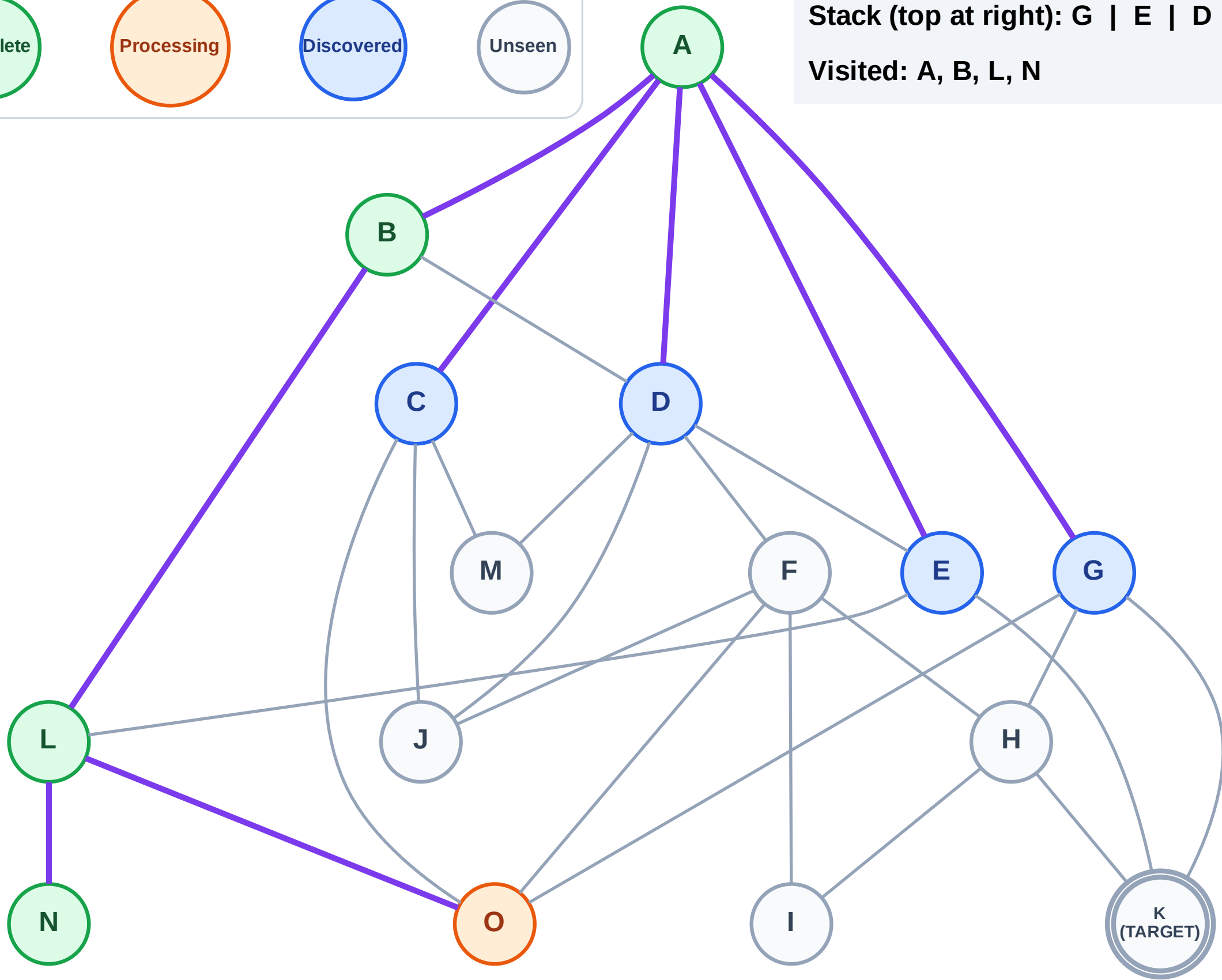
Step 10: Process node O

Legend



Stack (top at right): G | E | D | C

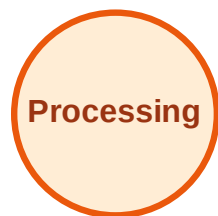
Visited: A, B, L, N



DFS Traversal

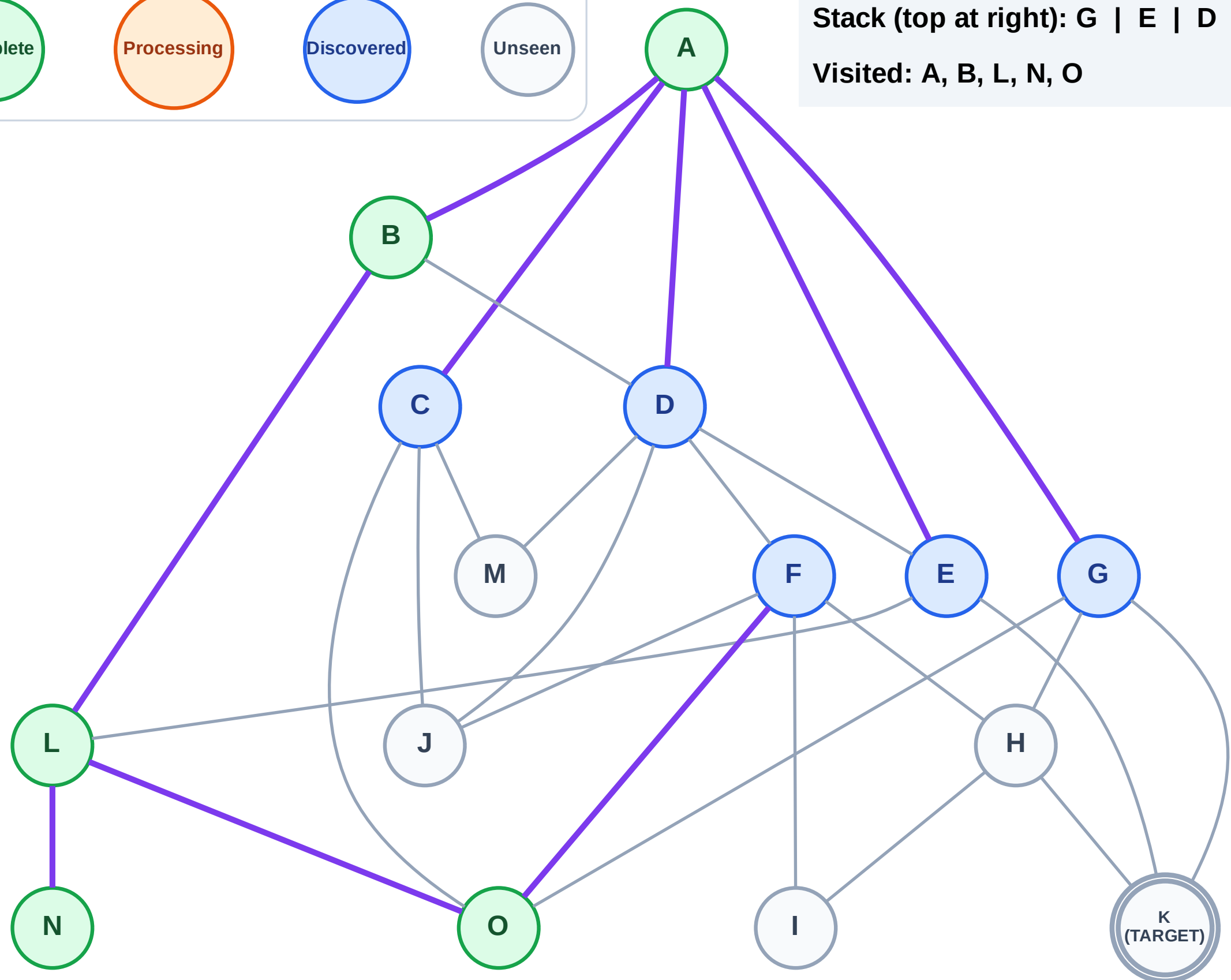
Step 11: Discover F from O

Legend



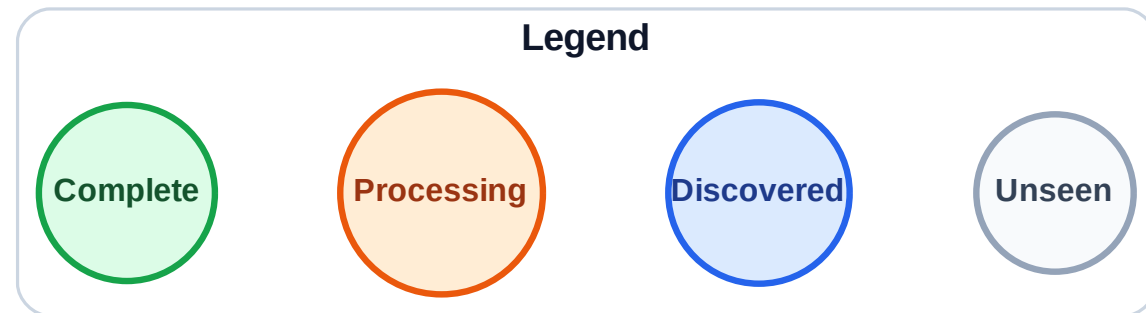
Stack (top at right): G | E | D | C | F

Visited: A, B, L, N, O



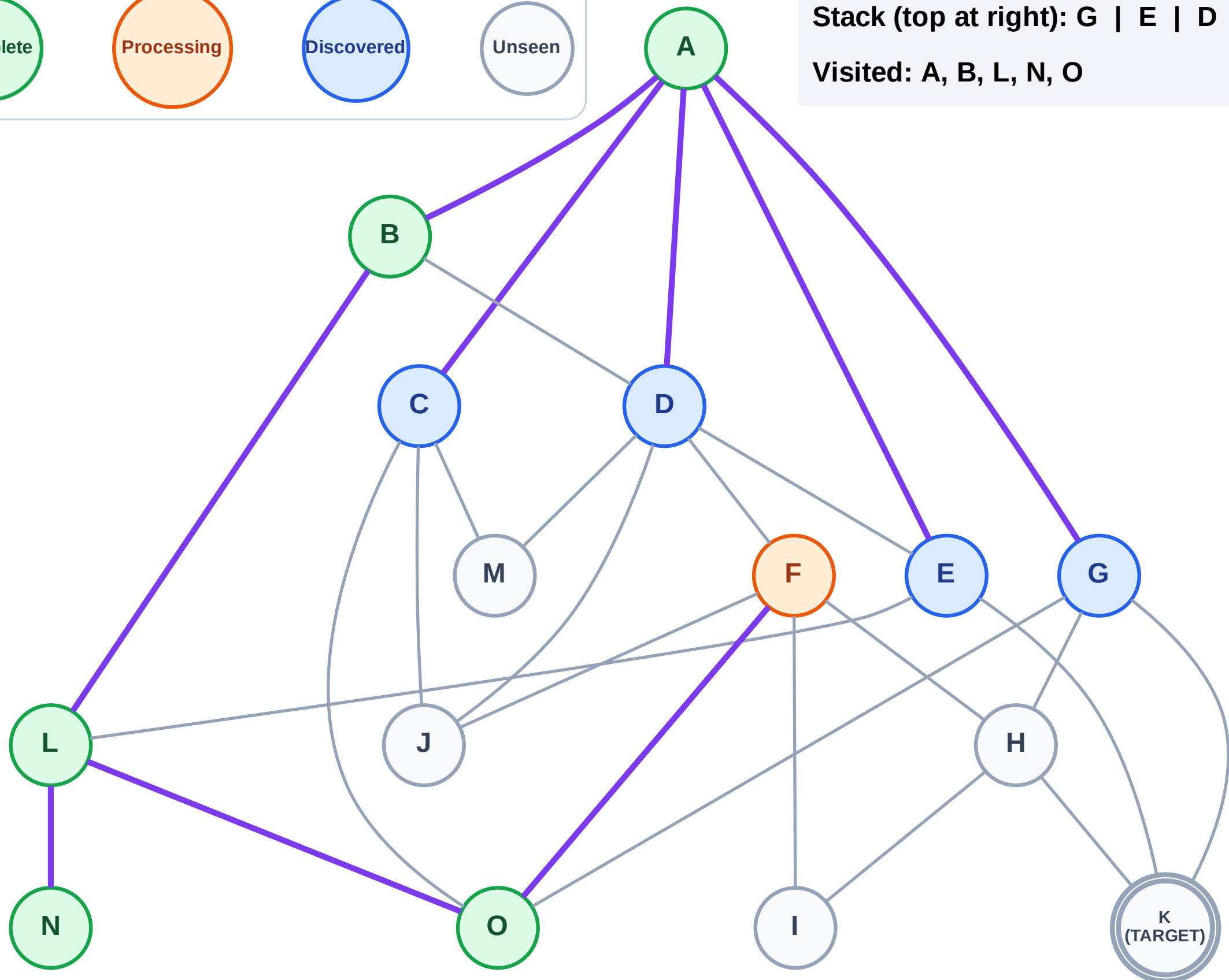
Step 12: Process node F

Step 12: Process node F



Stack (top at right): G | E | D | C

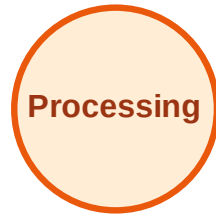
Visited: A, B, L, N, O



DFS Traversal

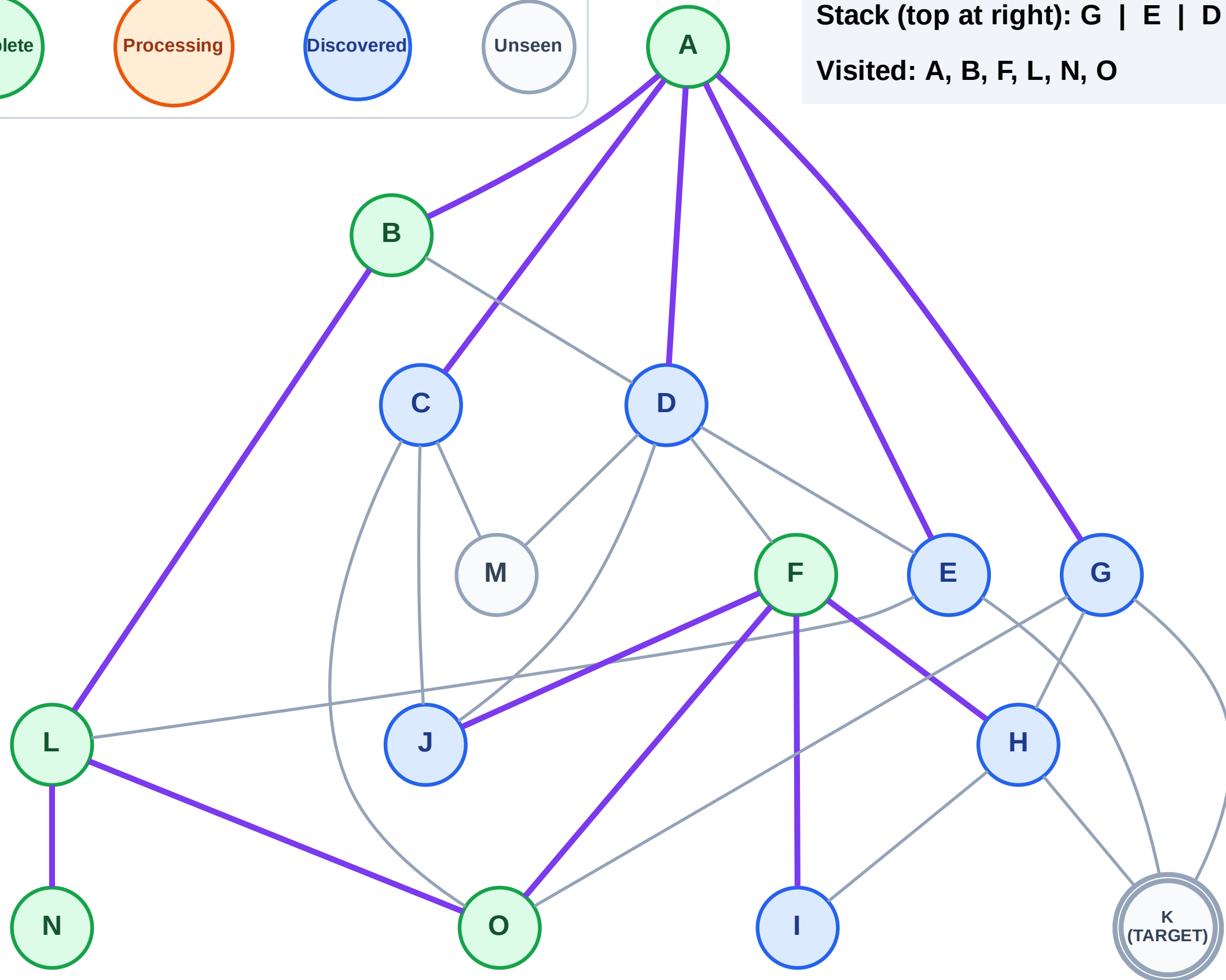
Step 13: Discover J, I, H from F

Legend



Stack (top at right): G | E | D | C | J | I | H

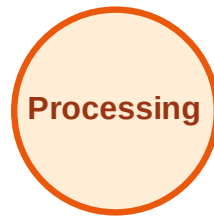
Visited: A, B, F, L, N, O



DFS Traversal

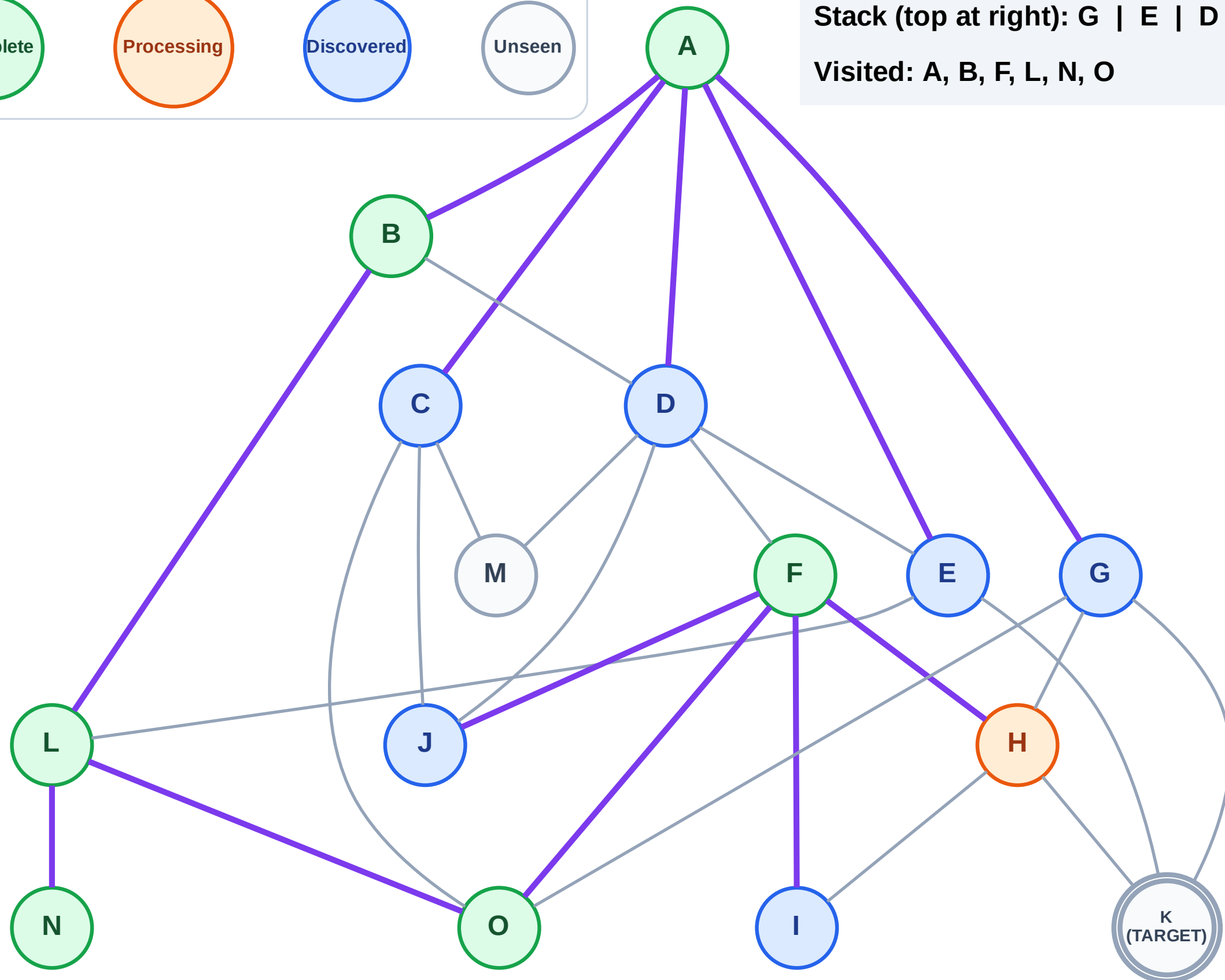
Step 14: Process node H

Legend



Stack (top at right): G | E | D | C | J | I

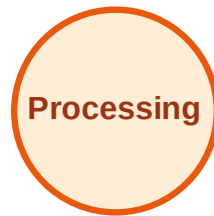
Visited: A, B, F, L, N, O



DFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (6 hops): A → B → L → O → F → H → K

Legend



Stack (top at right): G | E | D | C | J | I

Visited: A, B, F, H, K, L, N, O

